ECHO: The Signal That Was Never Sent

Entangled Minds: The Quantathon — Case Report

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Mandatory Modules: 01, 03 and 05
Stretch Module: 06 — The Impossible Claim
Case File: QX-ECHO-2026

1. Final Theory and Verdict

My verdict is that the ECHO experiments do not demonstrate faster-than-light communication. My investigation suggests that Dr. Zara Mahmud may have interpreted quantum correlations as evidence that entanglement could transmit information instantly. However, quantum entanglement creates correlated measurement outcomes; it does not provide a controllable channel for sending messages faster than light.

My confidence is high in this scientific conclusion because the simulated experiments demonstrate correlations and the need for classical information in the teleportation protocol. However, I have low confidence in any specific explanation for Zara's disappearance because the notebook cannot identify a responsible person or establish what happened in the laboratory.

2. Evidence and Clues

Clue 1 — Predictable Quantum Operations (Module 01): The X gate consistently produced `1`, while applying H twice returned the qubit to `0`. A single H gate produced approximately balanced outcomes. These results demonstrate that quantum circuits can create both deterministic and probabilistic measurement outcomes.

Clue 2 — Correlated Observers (Module 03): The Bell-pair circuit produced matching outcomes in both the Z and X bases. The CHSH test returned `S = 2.842`, exceeding the classical local-hidden-variable bound of 2 under the test's assumptions. This supports quantum correlations but does not establish faster-than-light communication.

Clue 3 — Message Recovery and Teleportation (Modules 05 and 06): Majority voting recovered the message `ECHO` from 32 bit triples. In teleportation, the receiver recovered the tested $|+\rangle$ state when classical correction information was used. Without corrections, the receiver's outcomes were approximately balanced. This demonstrates why shared entanglement alone is insufficient to complete the tested teleportation protocol.

3. Evidence Board

Category	Finding

FACT	Module 01 produced deterministic results and approximately balanced H-gate measurements.
FACT	Module 03 produced Bell correlations and a CHSH estimate of `2.842`.
FACT	Module 05 decoded the supplied bit triples as `ECHO`.
FACT	Module 06 recovered the tested state with classical corrections.
CLAIM	Zara believed ECHO could transmit information faster than light.
INFERENCE	The results support quantum correlations, not a controllable faster-than-light communication channel.
OPEN QUESTION	Who configured the final experiment, and what happened to Zara? The simulations cannot answer these questions.

4. Experiments and Results

Module 01 — The Gatekeeper: I tested X, H, H–H and a three-qubit circuit targeting `101`. The X circuit produced 1,000 ones; the H circuit produced 517 zeros and 483 ones; H–H produced 1,000 zeros; and the target circuit produced `101` in all 1,000 shots.

What it shows: Quantum gates can create predictable outcomes, superposition and reversible operations.

Module 03 — The Two Observers: I prepared a Bell pair using H and CNOT gates. The Z-basis measurements produced 992 outcomes of `00` and 1,008 of `11`. The X-basis measurements produced 995 outcomes of `00` and 1,005 of `11`. The CHSH calculation returned `S = 2.842 ± 0.0223`.

What it shows: The correlations are consistent with quantum entanglement and challenge local-hidden-variable explanations under the CHSH assumptions. Matching outcomes alone, however, do not prove communication.

Module 05 — The Damaged Message: I grouped the supplied 96 bits into 32 triples and applied majority voting. The decoder successfully recovered `ECHO`. The failure test also showed that one flipped bit can be corrected, while two or more flips can produce an incorrect decoded bit.

What it shows: A classical three-bit repetition code can recover information from limited corruption, but it cannot guarantee correct decoding when most bits in a triple are flipped.

Module 06 — The Impossible Claim (Stretch): I tested teleportation using the input state $|+\rangle$. With classical feed-forward, the receiver's X-basis verification produced 2,000 zeros. Without classical corrections, the results were 974 zeros and 1,026 ones.

What it shows: For this tested input, classical correction information is necessary to recover the intended receiver state. Entanglement alone is insufficient to complete the protocol.

5. Interpretation and Alternative Explanation

The combined results do not support ECHO's proposed faster-than-light communication mechanism. Module 03 demonstrates quantum correlations, while Module 06 demonstrates the importance of classical information in teleportation.

A plausible alternative explanation is that Zara performed a valid quantum experiment but misunderstood correlated measurement outcomes as evidence of faster-than-light signalling. I prefer this explanation because it is consistent with standard quantum mechanics. However, experimental error, altered records or deliberate intervention cannot be ruled out from the notebook evidence alone.

6. Next Experiment and Limitations

My next experiment would independently verify the CHSH circuit, measurement settings and outcome mappings, then repeat the measurements with more shots to estimate statistical uncertainty. I would also test teleportation with several input states, comparing receiver measurements with and without classical correction information.

A major limitation is that these results come from an ideal simulator, not physical quantum hardware. Finite measurement counts introduce statistical variation. The teleportation experiment tests only the specific input state $|+\rangle$, not every possible unknown quantum state. Finally, the simulations cannot establish what happened to Zara or identify a culprit.

7. References and AI Assistance

1. *ECHO: The Signal That Was Never Sent* — Entangled Minds: The Quantathon Participant Manual, Qubits Qrew, NED University of Engineering & Technology.
2. *Code of Conduct — Entangled Minds: The Quantathon*, supplied by the organizers.
3. IBM Quantum Learning: https://learning.quantum.ibm.com/
4. Qiskit Documentation: https://docs.quantum.ibm.com/

AI Assistance: OpenAI ChatGPT was used to help review the notebook, clarify quantum-computing concepts and organize the report. The experiments were executed using Qiskit Aer, and the reported numerical results were taken from the notebook outputs.

Final Conclusion: The investigation supports quantum correlations and the importance of classical communication in teleportation, but it does not demonstrate faster-than-light communication. The circumstances surrounding Zara's disappearance remain unresolved.